CS401p – Assembly Language

Assignment No. 1

Student ID: bc230424977

Name: QALB-e-MOMIN

Problem Statement No. 1:

You are required to write an assembly language program to perform the following tasks:

a) Store the numeric part of your VUID in an array of numbers in memory.
b) Find the smallest odd number of your VUID and store it in memory.
c) Add each numeric digit of your VUID to the smallest odd number (computed in part b) and store it in a 2nd array.

Note: You must use **your own VUID**. For example, if your VUID is BC260412345, the numeric part will be 260412345. Otherwise, zero marks will be awarded.

Submission details:

Following are required in a single MS-Word document.

a) Assembly language program.
b) Screenshot showing the results.

Solution:

➢ Code:

```
[org 0x0100]
jmp start

vuid db 2, 3, 0, 4, 2, 4, 9, 7, 7
smallest_odd db 0
result db 9

start:
mov si, 0
mov bl, 0
mov bh, 0

find_odd:
mov al, [vuid + si]
test al, 1
```

```
jz not_odd
cmp bh, 0
je first_odd_found
cmp al, bl
jae not_odd
mov bl, al
jmp not_odd

first_odd_found:
mov bl, al
mov bh, 1

not_odd:
inc si
cmp si, 9
jl find_odd

cmp bh, 0
jne store_smallest
mov bl, 0

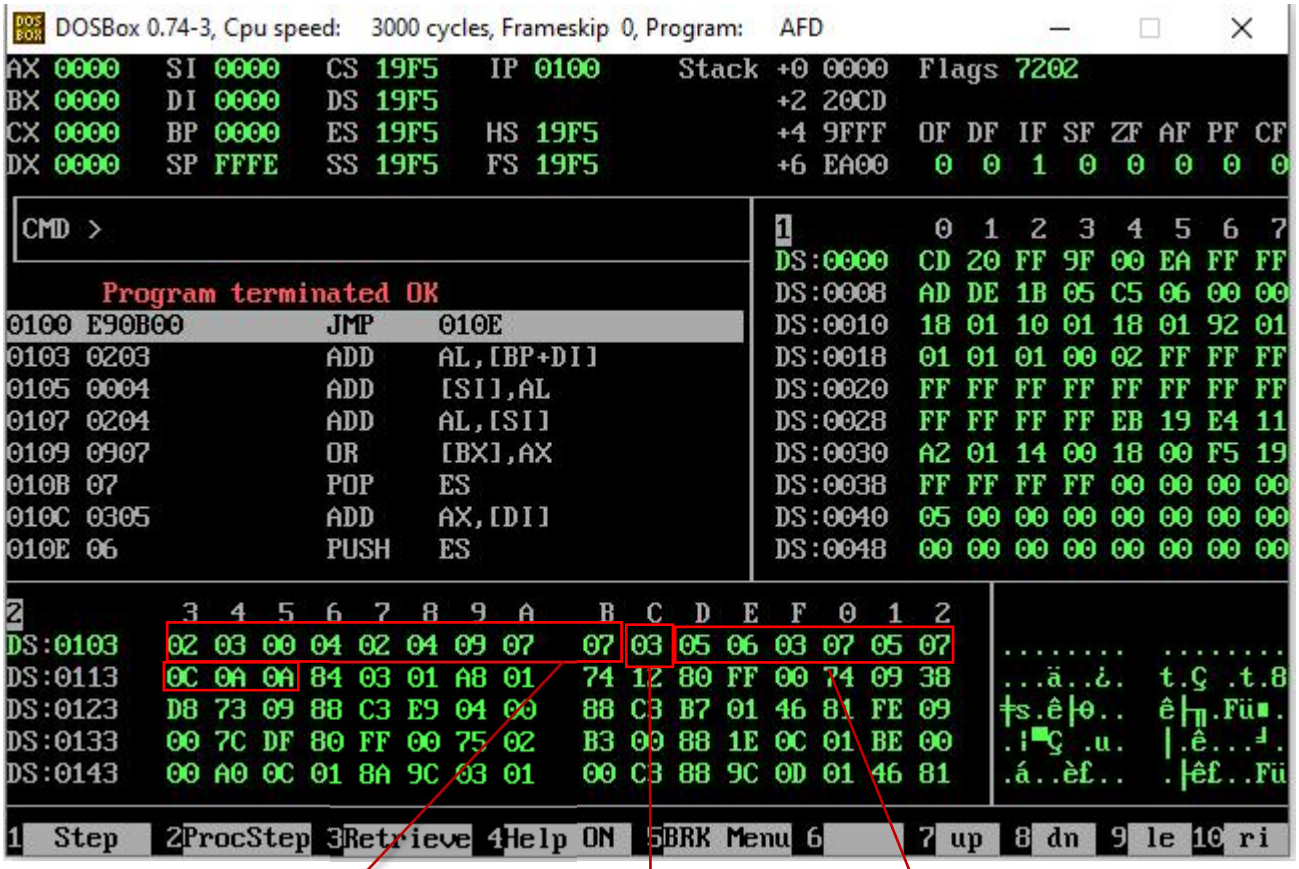
store_smallest:
mov [smallest_odd], bl

mov si, 0
mov al, [smallest_odd]

add_loop:
mov bl, [vuid + si]
add bl, al
mov [result + si], bl
inc si
cmp si, 9
jl add_loop

mov ax, 4c00h
int 21h
```

Screenshoot:



Original VUID **Smallest Odd Number** **VUID After Addition**